Safe System of work

Training and Assessment Requirements

Both SPRAT and IRATA maintain their own training, assessment, and competency requirements for each certification level. While the two systems are very similar in philosophy and work practices, there are differences in specific techniques, maneuver requirements, terminology, and assessment criteria.

Because of these differences, this section is intended to explain the various rope access maneuvers, identify which certification system they apply to, and clarify the competency level required within each system. This allows candidates and technicians to better understand the practical and theoretical expectations for both SPRAT and IRATA certifications.

The similarities between the two systems can sometimes create confusion, even for experienced rope access technicians. For this reason, this section is designed to serve as a practical reference guide throughout the training process.

At the end of this section, assessment checklists for both SPRAT and IRATA are provided to help clarify the skills, maneuvers, and competencies that candidates are expected to demonstrate successfully at each certification level.



Safe System of Work

Each certification system is supported by two primary governing documents that provide guidance for employers to develop, implement, and maintain a safe system of work, while also establishing the requirements for technician training and certification. In addition, IRATA International requires all member companies to undergo and successfully complete regular independent third-party audits to ensure compliance with international safety and operational standards.

SPRAT

- ☑ Safe Practices for Rope Access Work
- ☑ Certification Requirements for Rope Access Work

IRATA

- ☑ International Code of Practice (ICOP)
- ☑ Training, Assessment, and Certification Scheme (TACS)
- ☑ Audited Member company members and their operating procedures

SPRAT

- Candidates must demonstrate an understanding of their duties and responsibilities as written in the Safe Practices
- Candidates shall have a working understanding of the employer's rope access program and all applicable policy and procedures
- Candidates shall demonstrate the awareness of what constitutes appropriate supervision on the worksite
- Candidates shall demonstrate an understanding of the skills and knowledge required by Certification Requirements for Rope Access Work and Safe Practices for Rope Access Work

IRATA

- Candidates shall show awareness of the three elements listed above and the role of each in forming a safe system of work.
- Candidates shall show awareness of the ICOP's requirement that all rope access work is carried out within the management framework of an audited member company
- Candidates shall show awareness of the ICOP's requirement for supervision by a Level 3 rope access safety supervisor.
- Candidates shall show awareness of the levels of rope access technician competence, the role of logbooks, and the requirements for revalidation and progression between levels.

Jurisdictional Requirements

In many countries, government and regulatory authorities establish workplace health and safety requirements that apply to rope access operations. All rope access technicians and candidates must demonstrate an understanding of the jurisdictional regulations applicable to their workplace, including local laws, industry standards, and company procedures. Technicians must also understand how the safe system of work established by SPRAT and IRATA aligns with and is influenced by these legal and regulatory requirements.

SPRAT

- ☑ Level III candidates shall have sufficient knowledge of current regulations that apply, directly or indirectly, to rope access and working at height, so as to ensure compliance by the employees being supervised

IRATA

- ☑ Level 3 candidates shall demonstrate familiarity with the legal framework of their operating country, including any regulations that may be relevant to rope access activities.

Risk Assessment

Risk assessment is the process of carefully identifying hazards and evaluating potential risks that could cause injury, damage, or harm to personnel, equipment, or the environment. It also involves determining the appropriate control measures and precautions necessary to reduce risks to an acceptable level before work begins. Rope access technicians are expected to understand the basic principles of hazard identification, risk awareness, and safe working practices while operating at height. However, comprehensive risk assessment training is not included as part of this Level 1 curriculum.

SPRAT

- ☑ Level I candidates must be able to demonstrate an understanding of the employer's safety management program, relevant policies, work permits, work zones, and job safety analysis. Candidates should also be aware of course site hazards and emergency procedures.
- ☑ Level II candidates must be able to demonstrate an understanding of the employer's safety management program, relevant policies, work permits, work zones and job safety analysis.
- ☑ Level III candidates must be able to carry out the employer's safety management program including writing a job safety analysis.

IRATA

- ☑ All candidates shall show awareness of the role of risk assessment within the process of planning and managing rope access activities.
- ☑ Level 3 candidates shall demonstrate understanding of the risk assessment process, and shall be capable of identifying hazards that may affect rope access activities, e.g. falls from height, weather, falling tools and equipment, unusual loads, rope stretch, sharp edges, anchor failure, equipment failure, emergency evacuation, rescue provision for stranded/injured rope access technicians, etc.

Selection of Access Method

SPRAT

- Level III candidates shall have sufficient knowledge of current regulations that apply, directly or indirectly, to rope access and working at height, so as to ensure compliance by the employees being supervised.
- Level III candidates shall specify the appropriate rope access equipment, systems, and system components, and supervise their installation, use and inspection.

IRATA

- All candidates shall show awareness of the requirement to confirm that rope access is a suitable method for the task.
- Level 3 candidates shall demonstrate their understanding by assessing the suitability of rope access for a variety of scenarios.
- Level 3 candidates shall be capable of identifying common situations where other access methods such as mobile elevating work platforms (MEWPS), or using work restraint or fall arrest systems would be more appropriate.

Candidates should be able to identify the most appropriate method of working at height based upon a pre-job hazard analysis.

Selection of Personnel and Competence

All rope access candidates must demonstrate an understanding of the requirements for supervision, emergency response, and rescue provisions under both the SPRAT and IRATA systems. Technicians are required to recognize the importance of working within a controlled and supervised environment to ensure safe rope access operations.

Candidates must also understand the limitations of their certification level, including the scope of work they are permitted to perform, the extent of their rescue responsibilities, and the level of supervision required while carrying out rope access activities. Level 1 technicians are expected to work under the direct supervision of a qualified Level 3 Supervisor and must not undertake tasks or responsibilities beyond their training, competence, or authorization. Awareness of personal limitations and adherence to established safe work procedures are essential components of professional rope access practice.

SPRAT

- No additional specific training required

IRATA

- Level 3 candidates shall be capable of specifying a suitable team, including numbers of rope access technicians, levels of competence and roles, for a variety of scenarios.

SPRAT

- Level III candidates shall be able to carry out an employer's safety management program including comprehensive knowledge of a job safety analysis as stated in Safe Practices for Rope Access Work.

IRATA

- Level 3 candidates shall be capable of implementing a safety method statement, including briefing their team. They shall be capable of identifying when it may be necessary to revise a safety method statement and the procedure to be followed when doing so.

Method Statement

A method statement is a plan for effectively and safely conducting the assigned work. Based upon the safe system of work and the risk assessment, it states the sequence of procedures needed to complete work. It should incorporate a rescue plan covering all foreseen likelihoods. All candidates shall show an understanding of role of the method statement and the importance of following it.

SPRAT

- Access zone — the area in which people are at risk of falling such as on-rope or near a working edge. This area requires protective measures such as verbal warnings, signs, barriers, safety lines, or other devices designed to prevent or arrest a fall.
- Hazard zone — Any area where a person may be at risk as a result of the work being performed.

IRATA

- Level 2 and 3 candidates shall demonstrate an understanding of the process and requirements for creating and maintaining exclusion zones, including permit to work systems, size requirements, types of barrier and policing.

Exclusion Zones

Every effort should be made to protect workers and the general public from hazards while working at height. Exclusion zones are established as protection from falling objects and to prevent access to areas where falls from height could occur. Exclusion zones also prevent third parties from coming into contact with rope access operations. All candidates shall demonstrate an awareness of the types of exclusion zones and the need to create and maintain them.

Planning for Emergencies

SPRAT

- Level II candidates should have a working knowledge of rope access rescue procedures and emergency response considerations, including awareness of harness-induced suspension trauma and the importance of prompt and safe rescue practices.
- Level III candidates must demonstrate a strong understanding of rope access rescue procedures, emergency planning, and rescue management concepts, including harness-induced suspension trauma. Candidates are required to organize, supervise, and manage team rescue scenarios safely and effectively.

IRATA

- Level 3 rope access technicians shall be capable of preparing and implementing both evacuation and rescue plans.

All method statements for rope access operations should incorporate a plan for emergencies, including evacuation of the worksite and rescue of operatives. All candidates shall demonstrate awareness of their role in these plans, include basic first aid and suspension trauma.

EQUIPMENT USE AND SELECTION

SPRAT

- Level III candidates shall specify the appropriate rope access equipment, systems and system components.

IRATA

- Level 3 candidates shall be capable of selecting suitable items os equipment for a given work task and be able to identify situations where other equipment is more appropriate.

Equipment used in rope access operations should be selected based on its suitability for the intended purpose, compatibility with other system components, compliance with applicable regulations and standards, and the manufacturer's recommendations for use. All candidates must demonstrate an awareness of the principles of proper equipment selection and safe equipment use within rope access systems.

SPRAT

- Level I candidates must be able to demonstrate understanding of proper use, inspection and care of all equipment required for the technical skills. The candidate shall also understand the certification host's or employing company's equipment management program.
- Level II candidates must be able to demonstrate understanding of proper use, inspection, and care of all equipment required for the technical skills of a level II technician. The candidate should also understand the employer's equipment management program.
- Level III candidates must be able to demonstrate a thorough understanding of proper use, inspection, and care of all equipment required on a rope access work site. The candidate should be able to manage and carry out the employer's equipment management program.

IRATA

- All candidates shall demonstrate functional, visual and tactile pre-use checks of all personal equipment.
- Level 2 and 3 candidates shall demonstrate functional, visual and tactile pre-use checks of all ropes and rigging.
- Level 3 candidates shall demonstrate an understanding of appropriate marking and storage procedures for rope access equipment.
- Level 3 candidates shall make out an inspection report on the condition of damaged or work equipment provided.
- Level 3 candidates shall understand the limits of their level of competence in carrying out detailed inspections.

Care, Use and Maintenance of Equipment

Rope access equipment is exposed to conditions that may affect its safety and suitability for use, including wear and tear, abrasion, extreme temperatures, UV exposure, chemical contamination, and misuse. For this reason, all equipment must be inspected before use and undergo regular formal inspections.

Equipment should also be individually identifiable to ensure proper inspection and maintenance records are maintained.

All candidates must demonstrate an awareness of the factors that affect equipment integrity, as well as the manufacturer's recommendations regarding inspection, maintenance, lifespan, handling, use, and storage of equipment.

Anchor Selection

Rope access operations rely on a variety of anchors, including structural beams, eye-bolts, temporary ground anchors. Technicians responsible for selecting anchors must ensure they are competent, properly trained, and authorized to perform anchor assessment and selection in accordance with applicable standards and safe work procedures.

Demonstrate awareness of the minimum strength requirements for rope access anchors.						
SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Capable of inspecting and verifying the integrity of more complex anchors that may be built in the field by Level II and Level III candidates.						
SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Capable of selecting suitable structural elements to be used as rope access anchors.						
SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Basic Anchor Systems

At a minimum, a basic anchor system consists of a two-rope system, each with its' own anchor connection.

Demonstrate the rigging of a basic anchor for a two-rope system around a structural member.						
SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Demonstrate the rigging of a basic anchor for a two-rope system around a structural member while working at height.						
SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Load Sharing Anchors

Used when rigging ropes between two separate structural elements (horizontally or vertically).

Demonstrate the rigging of a small load sharing anchor system. Anchors are less than one meter apart (3.28 ft)						
SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Demonstrate the rigging of a wide load sharing anchor system. Anchors are more than one meter apart (3.28 ft). SPRAT requires a distance greater than 2 meters apart (6.56ft) IRATA requires the rigging to take place while at height.

SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Candidates show awareness of the issues of angle loading and potential swings associated with wide load sharing anchor systems.

SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Lowering Systems

In situations where top anchors are easily accessible and a clear vertical descent is available, rescue operations can be effectively managed using a pre-rigged releasable lowering system. This type of system allows the rescuer to safely lower a suspended casualty without the need for complex rope transfer procedures.

Candidates must demonstrate the operation and safe use of a simple pre-rigged rescue system designed for an unimpeded vertical lower, in accordance with approved rope access procedures and safe working practices.

SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Demonstrate rigging and lowering system.						
SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Cross Haul

A cross haul is used to move a load through a 3 dimensional course. All systems should utilize a back-up system. Generally a team exercise.

Demonstrate a cross haul of a load between two points.						
SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Team Exercise

Moving a casualty to a safe area may require a rescue system that incorporates several elements. Careful planning is required and this exercise should incorporate elements of planning, communication, and management. Candidates should be able to effectively create and implement a rescue plan.

Demonstrate a cross haul of a load between two points.						
SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Hauling and Lowering Through Knots

Raising and Lowering with Knots in the System:

Candidates must demonstrate the ability to safely raise and lower a casualty or load when knots are present in both the main and backup lines at a similar height, while operating either from a suspended position or from a platform. The casualty or load must be suspended at least 2 m below the knots, and the knots must be positioned at least 2 m below the anchorage system. The load or casualty shall be raised to the anchorage system and then returned safely to the original position.

Candidates may reach the anchorage system by any appropriate access method. However, no connections may be made to the casualty, load, or supporting two-rope system until the candidate has reached the anchorage area.

Knots used in the exercise must be located at approximately the same height and shall not be used as attachment points during the maneuver.

SPRAT			IRATA			
Level I	Level II	Level III	All	Level 1	Level 2	Level 3

Tensioned Lines

Ropes can be tensioned between two sets of anchors to facilitate near horizontal or diagonal movement. Additional working and safety lines are required to control movement. Candidates shall demonstrate an understanding of the high loads which could be placed on the anchors due to the creation of wide rigging angles. Ropes should be rigged with as little tension as possible. Proper back-up safety management is critical.

Demonstrate a cross haul of a load between two points. haul of a load between two points.oss haul of a load between two points.oss haul of a load between two points.

SPRAT			IRATA			
Level I	Level II	Level III	All	Level 1	Level 2	Level 3

Knots

Tying knots is a core element in working at height. Knots provide additional shock absorbing capacity when rigging anchor systems, and are commonly used in lanyard attachments, securing and joining ropes and to prevent descending off of ropes.

Candidates shall demonstrate the tying, tying, dressing and and setting of the he following knots and have an awareness of their strengths, applications and limitations						
Knot Category:	**End Knots**	**Knot Types:**	**Figure 8 on a Bight, Double Figure 8 on a Bight, Figure 9 on a Bight**			
SPRAT				**IRATA**		
Level I	Level II	**Level III**	all	**Level 1**	**Level 2**	**Level 3**
Knot Category:	Mid Knots	Knot Types:	**Alpine Butterfly**			
SPRAT				**IRATA**		
Level I	Level II	**Level III**	all	Level 1	Level 2	Level 3
Knot Category:	Stop Knot	Knot Types:	Barrel Knot			
SPRAT				**IRATA**		
Level I	Level II	**Level III**	all	Level 1	Level 2	Level 3
Knot Category:	Join Knot	Knot Types:	figure□eight knot			
SPRAT				**IRATA**		
Level I	Level II	Level III		**Level 1**	**Level 2**	**Level 3**
Knot Category:	Friction Hitch	Knot Types:	Prusik (Three-wrap)			
SPRAT				**IRATA**		
Level I	Level II	Level III	**all**	Level 1	Level 2	Level 3

Candidates shall demonstrate the tying, dressing and setting of the following knots, and have an awareness of their strengths, applications and limitations.							
Knot Category:	**Load Releasing Releasing Hitch**	**Knot Types:**	**Italian Hitch**				
SPRAT				**IRATA**			
Level I	**Level II**	**Level III**	**All**	**Level 1**	**Level 2**	**Level 3**	
Knot Category:	**Choking of Ropes and Slings**	**Knot Types:**					
SPRAT				**IRATA**			
Level I	**Level II**	**Level III**	**All**	**Level 1**	**Level 2**	**Level 3**	

Rope Protection

Care shall be taken to ensure that ropes are protected from hazards commonly found at the worksite. These hazards include sharp edges, abrasive surfaces, corrosive substances, and heat sources. Whenever possible, such hazards should be completely removed from the worksite.

If removal is not feasible, ropes must be rigged in a way that avoids contact with these hazards. As a final precaution, appropriate rope protectors should be installed to reduce the impact of the hazard to an acceptable level.

Personnel must demonstrate an awareness of the potential consequences of hazards affecting equipment, as well as the methods for removing, avoiding, containing, or reducing these hazards. They must also understand how movement (such as rope motion, load shifts, or environmental changes) may increase the risk posed by these hazards.						
SPRAT			**IRATA**			
Level I	**Level II**	**Level III**	**All**	**Level 1**	**Level 2**	**Level 3**

Demonstrate attaching wrap-around style rope protection while on rope. Separate protection for each rope is preferred						
SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Pull Through System

A pull through system allows ropes to be installed and retrieved remotely. Care must be taken to utilize the correct ropes and not the retrieval side. The candidate should identify the retrieval side to avoid any mistakes.

Demonstrate a method to retrieve ropes from a structural anchor after descent.						
SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Fall Factors

The fall factor is a numerical representation of the severity of a fall. Calculated by dividing the length of fall by the length of rope. The greater a fall factor, the greater the potential severity of the fall and corresponding impact forces generated.

Demonstrate understanding of fall factor and explain the method for calculating a fall factor.						
SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Rope Access Maneuvers

Back-up Devices

Although rarely used for arresting a fall, back-up devices are an important element of a safe work system. Correct use, maintenance and inspection of this device will be monitored throughout the training and certification course.

Demonstrate throughout the training and certification course the use of a back-up device in accordance to industry best practice, SkyTechAcademy risk assessment and the manufacturer's instructions.

SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Descent

The controlled descent of the technician is a core technique in rope access. Technicians must be able to control the rate of descent and stop as needed. Technicians should be capable of using alternate equipment to descend such as an ascender. Use of additional friction when needed such as when technician weight is over 100 kg (225 pounds) or in a rescue situation must be kept in mind.

Demonstrate attaching and function testing a descending device. Candidates shall demonstrate safe control of the standing or tail rope while descending. Candidates shall demonstrate stopping and securing (locking) the device.

SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Demonstrate ascending a rope at least 2 meters (6.56 ft) with a decent device.						
SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Ascent

Climbing a rope is a core technique to rope access. Technicians must demonstrate the alternating use of two ascending devices.

Demonstrate attaching and function testing ascending devices. Candidates shall demonstrate ascending a rope using ascending devices.						
SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Demonstrate descending a rope at least 2 meters (6.56') with an ascent device.						
SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Change-Over

Changing from decent mode to ascent mode (or vice-versa) are core techniques in rope access.

Demonstrate changing from ascent to descent mode and vice-versa.						
SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Deviations

Deviations allow the re-direction of ropes around obstructions, to allow for repositioning and to avoid hazards such as abrasions and hot surfaces. Candidates must avoid the potential for out of control swings.

Demonstrate passing a single anchor deviation (less than 20° from the anchors) in ascent and descent mode.						
SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Demonstrate passing a double anchor deviation (more than 20° from the anchors) in ascent and descent mode.						
SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Rope to Rope Transfer

Allows near horizontal movement. Candidates must avoid potential out of control swings through the use of four points of attachment when necessary. Candidates must be aware of how to use an appropriate knot as a secondary back-up.

Demonstrate transferring from one rope to another at least 2 meters (6.56') apart.						
SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Re-belay

A re-belay is a secondary set of anchors installed at any distance below the primary anchors. Candidates must avoid potential out of control swings through the use of four points of attachment when necessary.

Demonstrate passing a re-belay that is less than 1.5 meters (5') horizontally from the primary anchor						
SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Demonstrate passing a re-belay that is more than 2 meters (6.5') horizontally from the primary anchor.						
SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Passing Mid-Rope Obstructions

Knots are placed mid rope to isolate minor damaged sections, or to join two ropes. Candidates shall not use a knot, used to isolate a damaged section of rope, as a connection point.

Demonstrate passing mid-rope knots in both ascent and descent modes.						
SPRAT				**IRATA**		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

The candidate must be aware of how to use an appropriate knot as a secondary back-up.						
SPRAT				**IRATA**		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Demonstrate the ability to identify, isolate and pass minor damage in a rope by tying the appropriate knot.						
SPRAT				**IRATA**		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Negotiate Edge

Edges, such as roofs, platforms and parapet walls, present additional hazards such as contact hazard or awkward movement. Fall restraint methods must be used around unprotected edges. Appropriate edge protection must be used. Impact loads must be avoided.

Demonstrate passing a top edge obstruction in ascent and descent modes.						
SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Rope Protection

Rope protectors are installed mid rope to guard against minor abrasion hazards. Separate protectors should be used for each rope. Attention must be given to ensure candidates understand there are a variety of types of protection. The protection chosen should be appropriate to the hazard.

Demonstrate passing and replacing mid-rope protection in ascent or descent mode.						
SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Climbing with Fall Arrest Equipment

Fall arrest systems are used when it is not practical to use a personal fall protection system that prevents a fall. Candidates must understand the specific requirements of a fall arrest system including personal fall arrest devices, suitable anchors and clearance distance.

Demonstrate a vertical climb with the use of pre-installed temporary fall arrest system.						
SPRAT				IRATA		
Level I	Level II	Level III	All	Level 1	Level 2	Level 3

Demonstrate climbing using a shock absorbing Y-lanyard system, maintaining appropriate attachment at all times						
SPRAT				IRATA		
Level I	Level II	Level III	All	Level 1	Level 2	Level 3

Horizontal Aid Climb

Horizontal aid climbing is a technique used to move across the underside of a structure. Candidates must maintain two independent points of attachment at all times while minimizing fall distance and potential impact loading. Instructors and Candidates must be aware of the potential fall factor of the climber and take appropriate actions to ensure the fall factor < 1.

Demonstrate horizontal aid climbing using a series of fixed or movable anchors						
SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Vertical Aid Climb

Vertical aid climb is a technique used to climb a structure such as a steel lattice tower or a series of bolt anchors. Candidates must maintain two independent points of attachment at all times while minimizing fall distance and potential impact loading. Instructors and Candidates must be aware of the potential fall factor of the climber and take appropriate actions to ensure the fall factor < 1.

Demonstrate vertical aid climbing progressing primarily in suspension in an upward direction.						
SPRAT				IRATA		
Level I	Level II	Level III	all	Level 1	Level 2	Level 3

Rope Access Rescue Procedure

1. Purpose

This procedure defines the requirements and methodology for conducting safe and efficient rescue operations during rope access activities, in accordance with industry best practices (IRATA / SPRAT).

2. Scope

This procedure applies to all rope access personnel involved in work at height where a rescue may be required, including supervisors, technicians, and rescue personnel.

3. General Principles

3.1 Pre-Planning of Rescue Systems

- Pre-rigged rescue systems shall be the primary method when developing rescue plans.
- Where pre-rigged systems are impractical, an alternative rescue method involving a qualified technician intervention shall be established.

3.2 Availability of Equipment

- Rescue equipment shall be:
- Clearly defined in the rescue plan
- Pre-prepared and readily accessible

Wherever possible, an additional set of ropes and associated equipment shall be available for rescue purposes.

3.3 Rescue Hierarchy

- Pre-rigged rescue system
- Assisted rescue using additional equipment
- Rescue using casualty's equipment

4. Competency Requirements

- Rescue personnel shall be competent and trained in rope access rescue techniques.
- Candidates (trainees) shall demonstrate:

1. Practical use of the casualty's ropes and equipment
2. Understanding of rescue system limitations
3. Safe and controlled rescue execution

5. Rescue System Considerations

5.1 Use of Casualty Equipment

Although pre-planned systems are preferred, candidates shall be capable of performing rescues using the casualty's installed system.

5.2 Load Management

- Personnel must recognize that rescue systems may be subjected to loads exceeding normal single-person loads.
- This results in:

1. Reduced system safety factors
2. Increased equipment stress

6. Casualty Management

6.1 Suspension Intolerance Prevention

- Conscious casualties shall be instructed to:

1. Move lower extremities regularly
2. Maintain circulation to reduce risk of suspension intolerance

6.2 Positioning and Support

- Where necessary, the casualty may:

1. Use a work seat
2. Be repositioned to improve comfort and circulation

6.3 Handling

- Casualty handling must be:

1. Controlled
2. Smooth
3. Minimized to avoid further injury

7. Safe System of Work

7.1 Backup and Rope Management

- Maintain correct positioning of:

1. Backup devices at all times

- Prevent:

1. Rope entanglement
2. Rope-on-rope abrasion

7.2 Equipment Control

- Ensure:

1. Proper device operation under increased load
2. Continuous monitoring of system performance

8. Scene Assessment

Before initiating any rescue, personnel shall:

- Conduct a dynamic risk assessment, including:

1. Environmental hazards
2. Structural integrity
3. Equipment condition

- Confirm:

1. Anchor suitability
2. System stability

- Identify potential risks to:

1. Rescuer(s)
2. Casualty
3. Third parties

Rescue in Descent Mode

Most rope access work is conducted while in descent mode, making this rescue one of the more practical.

Demonstrate the rescue of an incapacitated incapacitated casualty in descent mode, from an adjacent set of ropes. Casualty can be approached from either above or below. SPRAT allows for a Level I rescue from an adjacent set of ropes or the casualty's ropes.						
SPRAT				**IRATA**		
Level I	**Level II**	**Level III**	**All**	**Level 1**	**Level 2**	**Level 3**

Demonstrate the rescue of an incapacitated casualty in descent mode, from the casualty's set of ropes. Casualty can be approached from either above or below.						
SPRAT				**IRATA**		
Level I	**Level II**	**Level III**	**All**	**Level 1**	**Level 2**	**Level 3**

Rescue in Ascent Mode

Rescues in ascent mode may be strenuous due to the necessity to lift the casualty in order to disengage the ascending device.

Demonstrate the rescue of an incapacitated casualty in ascent mode, either from the casualty's set set of ropes, or a separate set of ropes.						
SPRAT				**IRATA**		
Level I	**Level II**	**Level III**	**All**	**Level 1**	**Level 2**	**Level 3**

Level 1 Rescue Scenario (SPRAT)

Candidate shall perform the following during one combined exercise:

A pick off rescue of a casualty in ascent mode.

Approach casualty from an adjacent set of ropes

Change-over the casualty from ascent mode to descent mode

Candidate shall then perform a rescue from descent

Demonstrate the Level 1 Rescue Scenario						
SPRAT				**IRATA**		
Level I	**Level II**	**Level III**	**All**	**Level 1**	**Level 2**	**Level 3**

Passing a Deviation with a Casualty

Deviations may be rigged to position ropes or avoid obstructions. These may not be compatible with lowering systems, so the rescue plan needs to ensure sufficient competence within the team to deal with them.

Demonstrate descending with an incapacitated casualty through a single anchor deviation.						
SPRAT				**IRATA**		
Level I	**Level II**	**Level III**	**All**	**Level 1**	**Level 2**	**Level 3**

Demonstrate descending with an incapacitated casualty through a double anchor deviation.						
SPRAT				IRATA		
Level I	Level II	Level III	All	Level 1	Level 2	Level 3

Rope to Rope Transfer with a Casualty

A rope-to-rope transfer is used for near horizontal movement with a casualty. Such an exercise could be used to avoid obstructions or to return to an access point from underneath a structure. This rescue should be started on one set of ropes, not in the middle of the transfer. Candidates must demonstrate four points of attachment as necessary.

Demonstrate transferring, with an incapacitated casualty, from one set of ropes to another set of ropes rigged at least 3 meters (9.84 ft) apart.						
SPRAT				IRATA		
Level I	Level II	Level III	All	Level 1	Level 2	Level 3

Passing a Small Re-belay with a Casualty

Small re-belays are usually rigged to avoid obstructions or to reduce rope stretch on long drops. These may not be compatible with lowering systems, so the rescue plan needs to ensure sufficient competence within the team to deal with them.

Demonstrate a rescue of an incapacitated casualty past a small re-belay on descent. Re-anchors may not be offset by more than 1.5 meters (5 ft) from the anchors above.						
SPRAT				IRATA		
Level I	Level II	Level III	All	Level 1	Level 2	Level 3

Mid-Rope Transfer Rescue

A difficult rescue situation may arise where a casualty is suspended midway across a transfer maneuver. Candidates must demonstrate four points of attachment as necessary.

Demonstrate the rescue of an incapacitated casualty who is suspended: ☑ At any point in a wide (more than 3 meter, 10') rope to rope transfer or, ☑ At any point while crossing a wide (more than 2 meter, 6.56') re-belay						
SPRAT				IRATA		
Level I	Level II	Level III	All	Level 1	Level 2	Level 3

Passing Mid-Rope Knots with a Casualty

Mid-rope knots are placed either to extend ropes or to isolate small areas of damage. Knots may be at the same level or offset.

Demonstrate the rescue of an incapacitated casualty past a set of mid-rope knots.						
SPRAT			**IRATA**			
Level I	Level II	Level III	All	Level 1	Level 2	Level 3

Rescue from an Aid Climb

Where technicians are climbing directly on the structure, planning must consider rescue techniques. Team selection should take account of the time taken to reach and rescue an incapacitated casualty. Method selection should take into account belaying with communication.

Demonstrate the rescue of an incapacitated casualty who is suspended by work-positioning lanyards during an aid climb. The rescuer may remain on the structure and lower the casualty or rig ropes and descend with the casualty.						
SPRAT			IRATA			
Level I	Level II	Level III	All	Level 1	Level 2	Level 3

Rescue from an Aid Climb with Casualty on a Short Connection

Rescues from aid climb can be particularly complex when the casualty is on a short link connection, especially if the short connection consists of all-metal connections an only one connector is used. The short connection used in the assessment should be a two-connector attachment into a bolt anchor or a one-connector attachment into a short wire strop anchor.

Demonstrate the rescue of an incapacitated casualty from an aid climb, where the casualty is directly attached by a short connection and where there are no higher anchors.						
SPRAT			IRATA			
Level I	Level II	Level III	All	Level 1	Level 2	Level 3

Passing a Large Re-belay with a Casualty

Large re-belays are usually rigged to avoid obstructions or to reduce rope stretch on long drops. These may not be compatible with lowering systems, so the rescue plan needs to ensure sufficient competence within the team to deal with them.

SPRAT			IRATA			
Level I	Level II	Level III	All	Level 1	Level 2	Level 3

SPRAT Evaluation Matrix

The evaluation matrix to the right is the one used by SPRAT during your assessment. The skills are marked by white, grey or black boxes. If you look down the column for Level 1 for instance, you will see that the white boxes have numbers in them. These numbers relate to specific paragraphs in the Evaluation Guidelines document and are mandatory for all candidates. The grey boxes mean that these skills may be demonstrated based upon the discretion of the evaluator. The black boxes mean that the skill is not required by the level indicated. An 'A' in a box indicates that awareness of the skill is required for certification.

SPRAT
Society of Professional Rope Access Technicians

Technician Evaluation Form **Serial Number:**

Candidate Information:			
SPRAT Number:			
Last Name(s):			
First Name(s):			
Middle Name(s):			
Date of Birth:	Year:	Month:	Day:
E-mail:			
Phone:			
Street Address:			
City:			
State / Province:			
Zip / Postal:			
Country:			

Evaluation Session Information:	
Date:	
Location:	
Host:	
Evaluator:	
Evaluator Number:	
□ ID Verified □ Candidate Affidavit	□ Logbook Hours:

	I	II	III
Rope Access Program			
Roles and Responsibilities			
Equipment Use and Inspection			
Job Safety			
Management and Communication	■	■	
Team Scenario	■	■	
Individual Maneuvers			
Use of Back-up Devices			
Use of Descenders			
Use of Ascenders			
Change-overs			
Passing Knots		■	
Rope-to-Rope Transfer		■	
Deviation			
Re-anchor (>2 m)			
Negotiate Edge			
Rope and Sling Protection			
Horizontal Aid Climbing		■	
Vertical Aid Climbing	■		
Rescue			
Level 1 Rescue Scenario		■	■
Rescue through Knots	■		
Rescue through Deviation or Re-anchor	■	■	
Rescue from middle of Re-anchor or Rope-to-Rope	■	■	
Rescue from Horizontal Aid Climbing	■		
Rigging			
Knots: □ end □ join □ mid □ stop			
Hitches: □ prusik □ tied-off münter hitch	■		
Rigging and System Dynamics	■		
Anchorage Systems			
Hauling and Lowering System		■	■
Rope Access System Pre-rigged to Lower	■		
Retrievable Rope Systems	■		
Pitch Head Break in and Lower	■		■
Cross-Haul	■		
Hauling and Lowering through Knots	■	■	
Tensioned Rope Systems	■	■	
Written Test Score:	_____/40		

Evaluation Result: **Pass** **Fail** **Level Attained:** **N/A** **1** **2** **3**

Comments:

Evaluator Signature:

I hereby affirm that this candidate has performed as described on this form.

Candidate Signature:

I hereby affirm that I have been evaluated as described on this form and understand that I have the right to appeal the results of this evaluation session.

IRATA Assessor Checklist

An IRATA assessment will follow a scheme similar to the SPRAT evaluation in terms of white, grey and black boxes and therefore required skills.

IRATA ASSESSOR ASSESSMENT CHECKLIST	Assessment date:	
	Training centre:	

CANDIDATE NAME & LEVEL TO BE ASSESSED	L 1	L 2	L 3								

All required areas are to be assessed and marked in the appropriate box. P = Pass; Dis = Minor Discrepancy; F = Fail. Boxes marked 'A' are awareness only, use under direct supervision. See Technician Assessment and Certification Scheme for detailed marking criteria.

PLANNING & MANAGEMENT											
Level 1 Questions											
Level 2 Questions											
Level 3 Questions											
Hazard identification and risk assessment	A	A									
Job planning exercise (incl. access & emergency)											
EQUIPMENT											
Selection, care & maintenance of equipment	A	A									
Pre-use checking of equipment											
Assembly of equipment & buddy check											
Inspection of equipment	A	A									
RIGGING ***Assess all white boxes and a minimum of any 2 grey boxes***											
Anchor selection	A	A									
Knots and rope handling											
Basic anchor system											
Y-hangs (L1 small, L2 & L3 wide)											
Hazard avoidance and rope protection											
Re-anchors											
Deviations											
Retrievable rigging											
Work restraint lines											
Vertical fall arrest systems											
Tensioned lines											
RIGGING FOR RESCUE AND HAULING ***Assess all white boxes and a minimum of any 1 grey box***											
Lowering system											
Hauling systems	A										
Cross haul											
Complex rescue system (team rescue)											
ROPE MANOEUVRES ***Assess all white boxes and a minimum of any 5 grey boxes***											
Back-up devices											
Descent											
Ascent											
Changeovers											
Descent using ascending devices											
Ascent using descending device											
Deviations											
Rope-to-rope transfer Re-anchors (L1 –											
small, L2 & 3 – wide) Passing mid-rope											
knots Edge obstruction at the top Use											
of work seat (comfort seat)											
Passing mid-rope protection											
CLIMBING TECHNIQUES											
Horizontal aid climbing – fixed anchors											
Horizontal aid climbing – moveable anchors											
Vertical aid climbing											
Climbing with fall arrest equipment											
ROPE RESCUES ***Assess all white boxes and a minimum of any 2 grey boxes***											
Rescue from descent mode											
Rescue from ascent mode											
Passing a single-anchor deviation with a casualty											
Passing a double-anchor deviation with a casualty											
Rope-to-rope transfer with a casualty											
Passing a small re-anchor with a casualty											
Mid-transfer rescue											
Passing mid-rope knots with a casualty											
Use of tensioned ropes for rescue											
CLIMBING RESCUES ***Assess all white boxes and a minimum of any 1 grey box***											
Rescue from aid climb											
Rescue from fall arrest equipment											
Rescue from aid climb – short connection											
ADDITIONAL NOTES MADE DURING THE ASSESSMENT											

Rope Access Level 1 Technician

LEVEL 1 ACCESS REQUIREMENTS

1. Assessment Requirements
2. Ascender
3. Descender
4. Use of the Work Seat (IRATA)
5. Passing Knots
6. Rope to Rope Transfers
7. Deviations
8. Re-Belays (Small, IRATA; Large, SPRAT)
9. Negotiating and Edge and Installing Edge Protection
10. Climbing with Shock Absorbing Lanyards (IRATA)
11. Anchorage Systems
12. Hauling & Lowering Systems
13. Pick-off from Descent (IRATA)
14. Rescue Scenario (SPRAT)

Rope access training at Level 1 is structured around two essential components: theoretical knowledge and practical demonstration. Candidates are first required to understand the technical principles that form the foundation of safe rope access operations. This knowledge is assessed through a written examination and includes an understanding of equipment, safety factors, and general working principles. However, theoretical understanding alone is not sufficient; it must be supported by the ability to apply these concepts in real situations.

A key part of the practical assessment is knot tying. Candidates must demonstrate competence in several essential knots, including a mid-line knot such as:

- Alpine Butterfly
- End knots
- Double Figure of Eight
- Figure of Nine
- Joining knot
- Stopper knot

These knots must be tied correctly, securely dressed, and used appropriately, as they are critical for maintaining safety in rope systems.

In addition to technical skills, Level 1 candidates are expected to show an awareness of hazards and basic risk mitigation. This involves recognizing potential dangers in the work environment and understanding how they can be controlled or minimized. Equipment inspection plays an important role in this process, and candidates must demonstrate good habits by checking all equipment before use and ensuring it is safe and fit for purpose.

Another critical aspect of the assessment is the proper use of the safety backup device. This device is fundamental to rope access safety, and its correct handling is strictly evaluated. Candidates must avoid touching the body of the device during operation and instead allow it to travel freely along the rope as it is designed to do. Only the tether or connector may be handled when necessary.

Overall, Level 1 training emphasizes not only technical ability but also discipline, awareness, and adherence to safe working practices. Candidates must demonstrate that they can apply their knowledge effectively, maintain control of their equipment, and operate safely within established standards.

Assessment Requirements

The following Checklist shows what maneuvers are required for a Level 1 Rope Access Technician to perform during an assessment. Most of these skills will need to be demonstrated for either training scheme, others will be exclusive to one or the other scheme while a few will be exclusive to a single scheme. The remainder of this section pertains to the specific training that you'll receive to ensure that you can perform these tasks during assessment.

Skill	Req'd
Roles and Responsibilities	✓
Equipment Use and Inspection	✓
Job Safety	✓
Back-up Device Handling	✓
Ascender (Ascent / Descent)	✓
Descender (Ascent / Descent / Lock-off)	✓
Change-overs	✓
Pass Knots (Isolate Damaged Ropes)	✓
Rope to Rope Transfer	✓
Deviation (Redirect)	✓
Re-anchor (SPRAT > 6.6 feet, 2 meters IRATA < 5 feet, 1.5 meters)	✓
Negotiate Edge	✓
Rope and Sling Protection	✓
Horizontal Aid Climbing	✓
Level 1 Rescue Scenario	✓
Knots	✓
Anchorage Systems (<1 meter)	✓
Hauling and Lowering Systems	✓

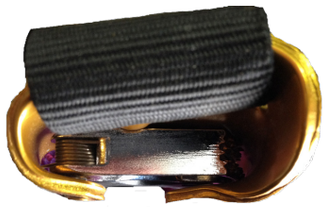
Ascender (Ascent/Descent)

In the last section you learned to ascend the rope. What wasn't covered was descent from the ascender. If you climb too high and need to re-position yourself down the rope a bit, you can do it without changing to the descender. If your descender fails while on the rope or if you, for some reason, forgot to bring a descender or dropped it, you can descend quite a far way using the ascender but it is a bit of repetition.

As the ascender is a toothed device, the load must be taken off of the Chest Ascender. To do this, place the Hand Ascender on the working rope, above the Chest Ascender as usual.

Slide it up so that if you were to stand up, all of the load is taken off of the Chest Ascender and transferred to the Hand Ascender. It is really useful to have the rope coming out of the chest ascender on the left side of your body. This will keep the rope further away from the tooth cam





Stand up in the foot loop and, using the hand not holding the Hand Ascender, press a finger onto the top of the toothed cam inside the Chest Ascender. While pressing with enough pressure to release the teeth from the sheath of the rope and holding the Hand Ascender handle, bend your legs to slowly let yourself down. Release the Chest Ascender cam and sit down completely to load the Chest Ascender. Move the Hand Ascender down the rope about the same amount of distance that you sat down and moved down the rope. Repeat as necessary, making sure you bring your safety back-up lower before you move further down.

Descender (Ascent / Descent / Lock)

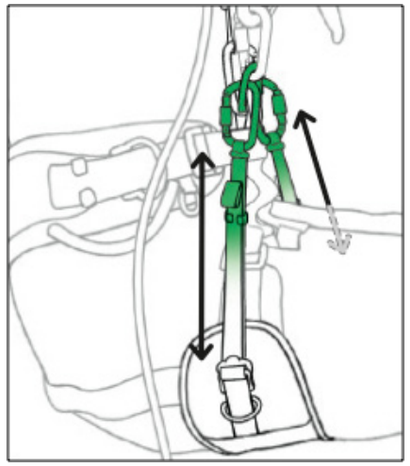
In the same way that you can descend on the ascender, you can ascend on the descender. If you've descended too far down, for example, you can ascend the rope by doing the following. This is pretty tough going without fairly new rope and good technique.



Sit on the descender to load it and lock it off. Place the Hand Ascender with the foot loop on the working rope and move it up the rope where it is reachable and usable. Unlock the descender and hold the trailing rope near the descender.

Using the foot loop, stand up fully and hold on t o t h e rope. While standing, the rope between the Hand Ascender and the descender is slack. With the hand holding the tail rope, pull out the slack. This is a timing thing, make sure you're pulling up on the standing end of the rope before you stand up too quickly. Sit down, reposition the Hand Ascender up the rope and repeat. Again, don't forget to reposition your back-up device.

Use of the Work Seat



A work seat is a means of being more comfortable while on the ropes. The work seat allows some of the pressure to be taken off of the legs allowing the Rope Access technician to work for long period of time more comfortably.



one on the left, and one on the right. A standard oval carabiner is attached to each strap and to the central attachment point of the Rope Access Harness. The straps are adjustable such that, if properly tightened, the seat acts as a load bearing member of the system. This takes some of the pressure off of the legs and is much more comfortable for prolonged time on the ropes.

Keep the seat underneath the buttocks or the upper thighs and not below the lower thighs at the bend of the knee. Having the work seat too low on the legs results in an unstable attitude and it is difficult to remain upright. The straps of the work seat should always be straight up and down.

Passing Knots

Sometimes you may find a section of rope that is damaged and want to isolate it or you need to connect two ropes together to continue descent or ascent. If either of these two instances occur, you'll have to pass the knot to either continue up or down. You will need two back-up safety devices to perform this maneuver correctly.

To pass the knot on the way up, ascend to just below the knot. You'll be approaching a fell factor 1, so manage your back-up properly. Attach the second back-up safety device above the knot



on the safety rope before your fall factor becomes close to a 1. Take the safety backup that is below the knot off of the rope and stow, and change-over from ascending to descending. If climbing in your descender, you'll want to position the descender to just below the knot.

Move the Hand Ascender above the knot with a moderate bit of rope between the Hand Ascender and the top of the knot in the rope. Change over from ascent to descent below the knot. Climb up in your descender until it is just below the knot.

Stand up and reposition your chest ascender above the knot. Close the Chest Ascender. This should be done as fast as possible. If the Chest Ascender is not on the rope and you do sit down, you will be sitting on the descender again and will have to do it again. Once you've successfully transferred the Chest Ascender to above the knot, sit down. Remove the descender from the working rope and stow. You may now ascend as normal.

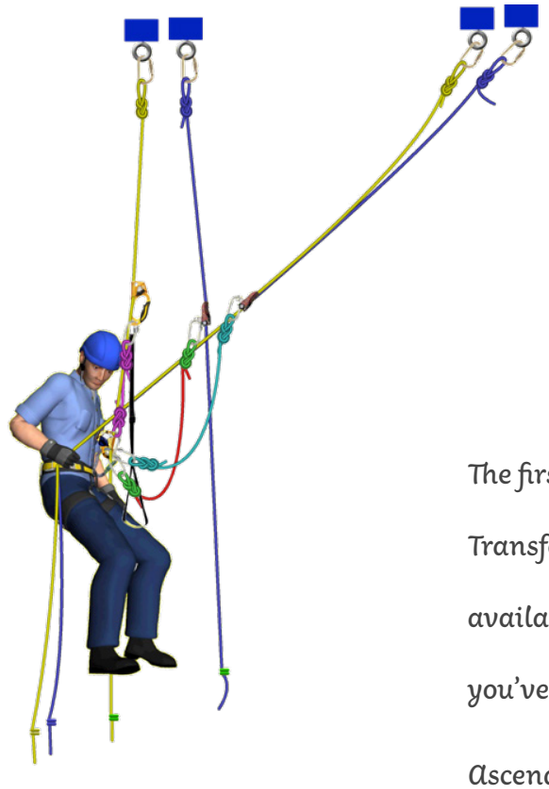
Descending and passing the knots is a similar exercise, with some variations from just reversing the steps:

Descend all of the way down to the knot. Change over to the chest ascender above the descender. Remove the descender from the Working Rope. Stand up in the foot loop and, using your finger to operate the chest ascender, descend down to the knot. You may have to do this several times to get both the Hand Ascender and Chest Ascender in position to remove and reattach below the knot. When at the knot, attach the descender below the knot. Remove the Chest Ascender and sit down to load the descender. Remove the Hand Ascender from the rope. Check your fall factor and descend if your fall factor is too high. Remove the back-up safety device that is above the knot, making sure you've installed a back-up under the knot on the Safety Rope.

Rope to Rope Transfer

From time to time, because of difficult access or other reasons, you may find it necessary to access the worksite by rope to rope transfer or a series of these maneuvers. Rope to Rope Transfers are not hard, there is, however, a few things to keep in mind:

1. Put something on the rope before you take something off; you must maintain 2 points of contact at all times.



2. Angle loading is serious business, keep your rope angles below 120°.

3.Performing Rope to Rope Transfers introduces a possible pendulum effect if something were to happen, so address this as a hazard in the risk assessment or JSA.

The first thing that you want to do when preparing for a Rope To Rope Transfer is to make sure that the ropes that you intend to transfer to are available to you. Tie an in-line figure of 8 in the ends of the ropes so that you've attached the two sets of ropes together.

Ascend (or descend) to the desired changeover location and, if you're in ascending mode, change to descending mode and lock the descender.

You'll need to use the knots you've tied at the ends of ropes. Pull the set you're on until you have the destination ropes in your hands.

Decide which of the transfer ropes will act as the safety back-up rope and place the second safety back-up device on it. Push it up as far as it will go without tensioning the cow's tail. Place the Hand Ascender on the other transfer rope and push it up as far as it will go.

Put the Chest Ascender onto the same rope as the Hand Ascender and pull all of the slack out of the transfer rope. Move the Hand Ascender up the transfer rope and step up on the foot loop to move up the transfer line. continue climbing until it becomes difficult. Check the positions of the two safety back-up devices and adjust up to keep them above the head.

Once there is tension on the system, you'll have to descend on the original lines to move horizontally toward the transfer ropes. This will lower the rope angle too, which is a good thing. Lock off the descender and, using the Hand Ascender and Chest Ascender, continue to ascend the transfer ropes.

Once you descend until your descender doesn't seem to work any longer, you are probably close to vertical on the transfer ropes. Check your back-up on the first set of ropes. Sometimes your descender will appear to be slack before you reach vertical. This happens because you've loaded your safety back-up. Do not remove your descender until you're sure you're under your destination anchors.

Pull slack out of the top of the descender to ensure that you are vertical. Since there is no load on the original rope system, you

should grab the ropes below all of the equipment on the lines, pull up until the ends are readily available to you and attach these to your harness in case you need them to transfer back later.

You may now remove the original descender and safety back-up device. If you need to ascend the transferred ropes, do so. If you need to descend to the work area, change over to descent mode and do so.

Single Deviations (Redirect)

A deviation is used to redirect the ropes in some manner, possibly away from an edge or around an edge and under a deck for instance. A deviation is made by passing the ropes through a carabiner that is attached to a sling of the correct length to position the ropes where they need to be. Of course the sling is connected to some structural member that will hold the load. Remember to consider angles of the rope so as not to load the deviation too much. Consideration should also be given to the swing radius should the deviation fail.

If a deviation is in place and you are below it, ascend to the deviation. Make sure you tie a slip knot in the ropes about 10' below the deviation (you'll use the knot to return to the deviation later). if you don't have an extra anchor with a carabiner attached, add a carabiner to the existing deviation anchor, making sure the carabiner is under the existing carabiner holding the ropes. Reach down below you and grab the ropes under all of your equipment.

Place these ropes into the empty carabiner making sure you're not twisting the ropes. Remove the working and safety ropes from the original deviation anchor. Grab the ropes under the new carabiner holding the ropes and gently let yourself swing away from the deviation. Be sure to keep your back-up as high as possible.

Descending past the deviation begins by descending to the deviation so that your main attachment point is even with the deviation. Lock off the descender. Pull your ropes under your equipment towards you until the knot rests directly under the deviation carabiner. Reach over and grab the knot, pulling the knot towards you. this action will tighten the ropes between your equipment and the deviation anchor. Continuing to pull will allow you to position yourself at the structure. Grab the original deviation carabiner (empty now) and attach to your working line above your descender.

You are now positioned with your descender below the deviation anchor and will not have a potential swing. Check your fall factor and descend a bit if necessary in order to maintain a low fall factor. Pull your backup device down the safety line until it is below the deviation anchor supporting you. Clip the rope above your safety line into the carabiner and check to make sure it is locked. Remove the carabiner which is on the ropes below your equipment and descend to the ground.

Double Deviations (IRATA)

Double deviations are used if the failure of a deviation would lead to catastrophic results such as high swings into rotating equipment or hot piping.

The approach is similar to a single deviation, but has significant differences, Ascend to the double deviation as with a single deviation. Once in position, tie a slip knot about 10' below the lower deviation. Attach a spare cow's tail to the anchor of the upper deviation. Attach a short link to the anchor of the lower deviation and connect the other end to the ventral attachment of your harness.

You can now release the ropes from the carabiners of the deviation, Be careful, though, you've got to pull yourself into the structure while doing this and then let yourself out so that the short link is keeping you close to the structure. now that you have the carabiners of the deviation free, lift the ropes below your equipment and insert those ropes into the carabiners, being careful not to twist or cross the ropes or deviation anchors.

Remove your cow's tail from the upper deviation anchor and pull yourself into the structure so that you can remove the short link from the lower deviation anchor. grab the ropes under the deviation carabiners before letting go of the structure. Pull those ropes tight and let go of the structure. Use the ropes in your hand to ease yourself out until you are under your upper anchors.

It is now safe to continue your ascent to the worksite. Make sure you change over to descent mode so it will be easier to rescue you if something bad happens.

When you're ready to go down through the deviation, just descend down until your descender is a bit lower than the bottom deviation's anchor. Use the ropes leading to the deviation to pull yourself closer to the deviation and grab the structure. attach the short link to the lower deviation anchor. If you find it too hard to connect the short link, let yourself back out and descend a bit more.

The short link should be attached to the ventral attachment of your harness. If you let go of the structure, you'll be supported by the short link. attach a spare cow's tail to the upper deviation's anchor and take the ropes out of the deviation carabiners. Pull yourself into the structure and attach the lower deviation carabiner to your working line above your descender.

Add the upper deviation carabiner to the working line above the lower deviation carabiner already attached to the working line. you'll probably have to pull yourself into the structure.
So, there are a lot of little steps to remember, but you're almost there. Your descender is below both deviation carabiners, now it's time to focus on your safety. Descend down just a bit so that you have an acceptable fall factor. Move your safety device so it's below the lower deviation carabiner and put the rope through both deviation carabiners.

Make sure the carabiners are closed and locked. Remove your extra cow's tail from the upper deviation anchor. Pull yourself into the structure and remove your short link from the lower deviation anchor. Descend down to the ground. You will most likely have to remove the knot you placed in the ropes, but that's easy.

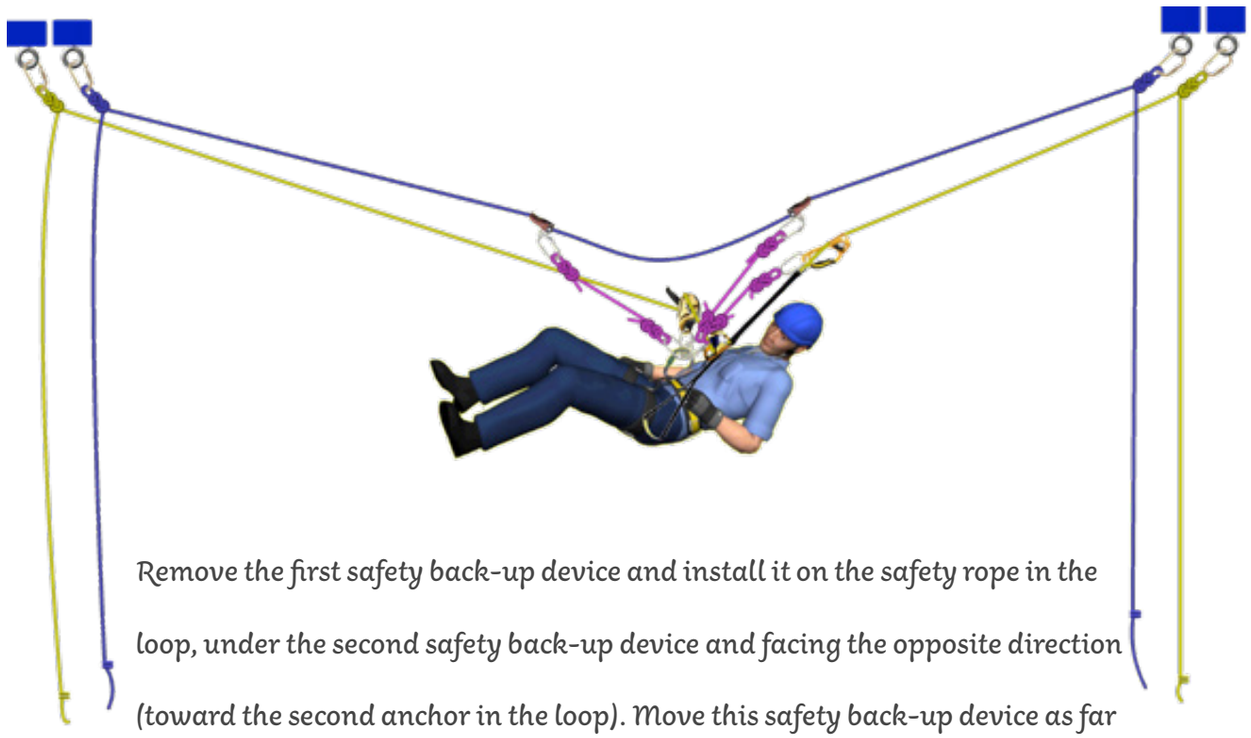
Passing Re-anchors

A re-anchor is when the direction of the ropes have to be changed and its not practical or possible to use deviations either to avoid damage to the rope or protection for the Technician. Some people call a re-anchor a re-belay or a loop. The thing about a loop is that it has a portion going down, and a portion going up. That predicates the use of both ascent and descent equipment and protection from a fall in both directions.

In reality, a passing a re-anchor is much like performing a Rope to Rope Transfer. If ascending, approach the intermediate anchor (the re-anchor) from the bottom and stop near the anchor. You'll commonly see two ropes coming from each of two knots. Place your extra safety back-up device on the extra rope coming from the same knot as the rope that your original back-up safety is on (we call that "making twins"). Make sure that your second backup is pointing towards the knot, in the same direction as the

original back-up. Place your descender on the other loop rope (on the rope that comes out of the knot your Chest Ascender is opposite), pull all of the slack out and lock off the descender.

Step up in the foot-loop and release the chest ascender from the rope going to the ground. This will sit you on the descender.



Remove the first safety back-up device and install it on the safety rope in the loop, under the second safety back-up device and facing the opposite direction (toward the second anchor in the loop). Move this safety back-up device as far forward (down the loop) as possible without tensioning the cow's tail. You'll have to pull plenty of slack through the second backup, creating a loop between the two safety devices. Place the hand ascender on the loop towards the distant anchor after pulling some slack to give you room to work. Now place the chest ascender on the same rope, in the same direction as the hand ascender, between the descender and the ascender.

Once all of the gear is in place, pull the slack through the Chest Ascender and Hand Ascender. This allows you to climb towards the far anchor putting more slack in the loops below you. Make sure you appropriately move the safety back-up device as you progress. Once your progress gets difficult, leave the ascending gear alone and unlock the descender and, using the safety backing up the descent device move the back-up down, descend a bit. Lock off the descender. Using the Chest Ascender and Hand Ascender, continue to move up the loop rope

making slack between the Chest Ascender and descender. Continue this operation until you are vertical remembering to move the safety back-up devices as necessary. Once you are vertical, perform another Rope to Rope Transfer (to descending gear) to descend down the new rope past the last anchor. This only works if there are down ropes. If there are no descending ropes, you'll have t o change over into the descender and reverse direction.

Negotiating an Edge and Installing Edge Protection

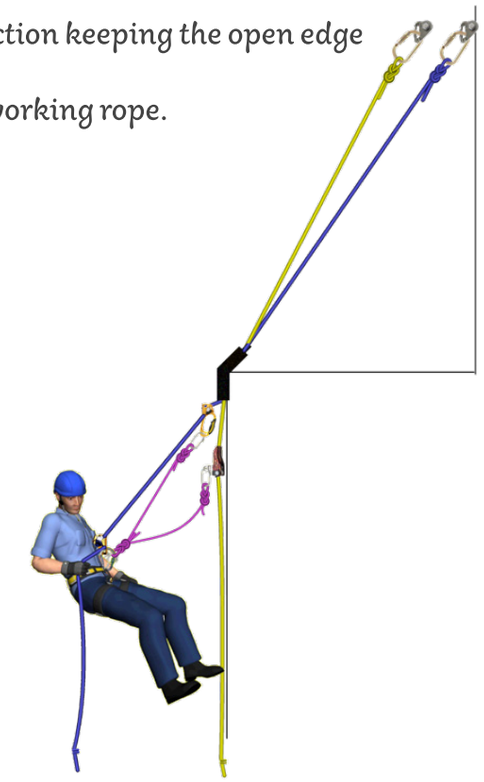
These two skills go together because part of negotiating an edge (going over an edge) is installing edge protection.

To negotiate the edge from below the edge, ascend to the rope protection at the edge. Open the rope protection on the safety rope, which is slack. Push the safety back-up past the edge making sure that the rope remains protected by the open edge protection.

If the rope protection is the non-opening type, use a knot passing technique.

Open the working rope edge protection keeping the open edge protection between the edge and working rope.

Ascend as far as possible



Change over to descent mode. The Hand Ascender will probably get near the edge and become very difficult to move. Remove the Hand Ascender from the working rope and move it up past the edge, placing it back on the working line.

Climb over the edge, connect a cow's tail to an independent anchor if necessary and disconnect all climbing equipment. Put the rope protection back into place.

If the edge is below you, while on the ledge, attach the safety back-up to the safety rope and the descender to the working rope. Take slack out of the descender and lock. Move to the edge, being very careful of the edge and position your safety back-up so that there is little slack in the cow's tail. When in position at the edge, sit down. Add additional slack in the working line above your descender so that the descender will clear the edge by about 2". There is stretch in the rope that will help position the descender over the edge later.

Attach your Hand Ascender, on the working rope, so that it is positioned over the top of the edge far enough so that it will not be loaded on the edge when you step into your foot-loop. Guys and gals, this can be difficult, so pay attention. Adjust your foot-loop so that it is at its shortest length. Put your foot into the foot-loop and step into it, rotating your body so that you are standing in the foot-loop over the edge.

Position the working line so that it is in the middle of the rope protector. Lean back and load your working line. Begin to close the rope protectors above the edge. Pay attention to the position of the back-up; you may have to move it down past the edge to reach the top of the rope protector. Ensure that you don't make a large fall factor but can still do up the protector. Descend down the rope a bit if your fall factor is too high. Unlock the descender and descend slowly, closing the rope protection as you descend, After the edge protection is in position, unlock the descender and continue the descent.

Climbing with Shock Absorbing Lanyards (IRATA)

There are times that you will not be using ropes to access the work site such as work on a gantry, under or along side a deck, etc. In these cases, you will be doing something called aid climbing or something called climbing with shock absorbing lanyards.

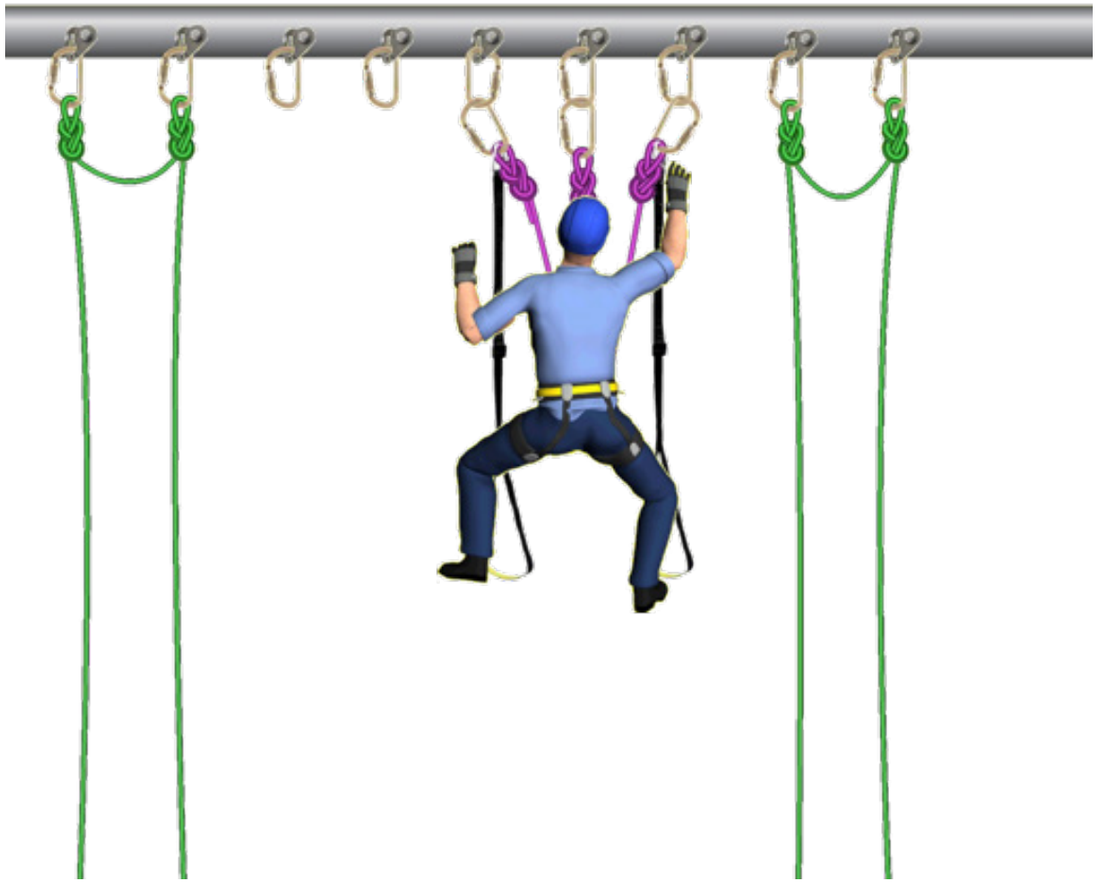
Climbing with Shock Absorbing Lanyards means, at this level, an awareness of the rules for shock absorbing lanyards (ANSI Z359) and how to use them. For this purpose, a shock absorbing lanyard is a device with sewn together sections that elongate as the stitching breaks while arresting the fall. The lanyard is generally a one-piece system that is shaped as a "Y". The leg of the "Y" connects to the dorsal "D" ring. The "Y" arms are positioned on, say, ladder rungs or the like and they are moved up as the technician moves up the ladder. Note that you must keep three of your four supports on the ladder or structure while you are climbing; two arms and a leg or an arm and both legs.

The other type of climbing on shock absorbing lanyards involves something called Aid Climbing.

Aid Climbing (Fixed)

So, this seems like a difficult thing to do, but it's really easy if you get organized first. Grab an extra foot-loop or etrier. You have a left, center and right cow's tail. The climb begins on a typical two rope system. You'll use the left cow's tail for your safety back-up and the right cow's tail for your Hand Ascender and foot-loop. The only thing on your center cow's tail should be a carabiner. Climb up to the point where the anchor slings to be traversed are reachable at a comfortable height. Change over to your descent gear.

While on the descent equipment, hook the center cow's tail into an anchor carabiner above your head. Place this cow's tail into your chest ascender and raise yourself up the cow's tail until you can comfortably reach the anchor. You'll probably end up close to the knot of the cow's tail. Remove the Hand Ascender from the right cow's tail and stow it on your harness using an extra carabiner. Attach that cow's tail to a second anchor on the structure. Descend down until you're fully supported on your center cow's tail. Remove the descender from the rope system. You are now in aid climb, but we need a third attachment point so



that you can move your cow's tails while being fully protected. Take the back-up safety device off of the safety rope and stow the back-up safety device. Attach your spare foot-loop to the carabiner on the end of the cow's tail. Hook it onto the same anchor as the center cow's tail.
So, you have three cow's tails connected to two anchors. Go ahead, count 'em. Your cow's tails should not be tangled. If you're moving to the right, your right cow's tail should be on your right side and your left cow's tail on your left. Take a moment to sort this out if you need to. Just remember to maintain two i ndepe nde n t po i n t s o f contact.

Key to this maneuver is to always have two independent points of contact at all times. This is accomplished by paying particular attention to the cow's tails with the etriers and moving the cow's tail connected to the Chest Ascender only when the other cow's tails are in position. A mistake often made in this exercise is that since there are so many connections and disconnections, screwing the carabiners completely is sometimes overlooked.
To move across the aid climb, attach the cow's tail in the direction of travel to the next anchor, move the farthest cow's tail to the same anchor as your center cow's tail, then step up into the foot-loops and move your center cow's tail to share the anchor of the cow's tail in the direction of travel. The rhythm should be: if you're moving to the right, R, L, C, R, L, C...if you're moving to the left, L, R, C, L, R, C....

Mobile Aid

Another type of aid climbing is using three anchor slings that are slung over a beam or pipe that the technician attaches to in a similar manner. This is called Mobile Aid. By sitting on the middle anchor sling and moving the two anchor slings with etriers, lateral movement is possible. When the technician stands in the etriers, the middle anchor sling may be moved and the sequence repeated.

Moving around an obstacle entails moving the leading anchor sling with an etrier around the obstruction, moving the center anchor sling to the obstacle and, while standing, moving the center carabiner to the anchor sling previously moved past the obstacle. The sling without any connection is then moved over the obstacle and placed past the first sling. You can then move the furthest cow's tail (and foot-loop) to this sling's carabiner. The remaining, trailing anchor sling, can now be moved in a similar fashion.

Pick-Off (Casualty Descending)

This is a rescue technique. We train people to always change over into the descending gear after ascending because the pick-off while the casualty is in the descending gear is the fastest and easiest way to rescue somebody. The first part of rescue is to make sure that any rescue that is to be undertaken can be performed safely; it's easier to save one person than it is to rescue a casualty and a second who was the rescuer.

Rescue begins with determining why the casualty is in the state that requires rescue. If the casualty can speak, ask why he or she needs help. These responses will help you to better plan the rescue. If the casualty cannot communicate, determine if the environment caused the person to become unconscious or if an injury caused the incapacitation. For instance, rescue of an individual exposed to hydrogen sulfide or nitrogen asphyxiation is not wise without full breathing air available. Were a rescuer to try a rescue without supplied air, there would be two casualties that would require rescue.



To begin the actual rescue (assuming that you've ensured that you will be safe in doing the rescue), connect your back-up safety device to the casualty's working rope and your ascenders to the casualty's safety rope. This will allow approximately equal loading of both ropes during the ascent.

Ascend to the casualty until you are level with the casualty. Place your spare back-up safety device

above the casualty's descender and push it all the way up. Remove your first back-up device from the rope.

Ascend past the casualty and change over into your descender. Make sure that when you sit down on the descender that you do so with the carabiner gate pointing so that the opening and the gate is facing you. Do this so that you are higher than the casualty. Lock off the descender. Make sure the ropes are not trapped between you and the casualty, move them to one side.

Using one of the casualty's cow's tails, establish a link between the two of you by clipping it into your ventral attachment. This establishes one point of contact to the casualty. Establish the second point of contact by connecting a short link, or two carabiners connected to each other. This should be accomplished by placing one side of the short-link to the sternal connection of the casualty and the other end to the back of the descender. It's better if you attach the sternal connection and pull the casualty up to make the attachment to the back of the descender and not the other way around. Note that if the casualty is using an I'D, position the short-link on the gold side not the back of the casualty I'D; you'll have to open his or her I'D and it's hard if the connection is on the handle side.

Using the casualty descender, lower the casualty onto your gear. Your descender carabiner now holds both loads, you and the casualty. Remove the casualty's descender from the rope. Remove the casualty back-up safety device from the rope. You might consider removing the casualty's safety back-up after establishing two points of contact to the casualty, depending upon the length of the casualty's cow's tail.

Place an extra carabiner on the carabiner connecting your descender to your harness, on the side of the descender facing you. Feed the slack end of the rope coming from the descender through this carabiner. This will act as extra friction while descending allowing you to control the descent. To use this extra friction, hold the rope UP, not to the side.

It's worthwhile to note here that you must maintain a small fall factor by keeping your back-up device as high as practical. Never allow the back-up device to be low enough to create a fall factor greater than .5.

Unlock your descender, pulling the slack end of the working line upwards so the carabiner you installed applies friction. Descend until you are near the ground maintaining that the casualty does not fully lie on the ground. It is imperative that all of the weight is not taken off of the casualty all at once as, if the casualty has been suspended without movement for a prolonged period of time, the sudden release of pressure could send too much blood surging through the arteries and cause additional injury. Slowly lower the casualty to the ground all the while moving the legs to restore circulation.

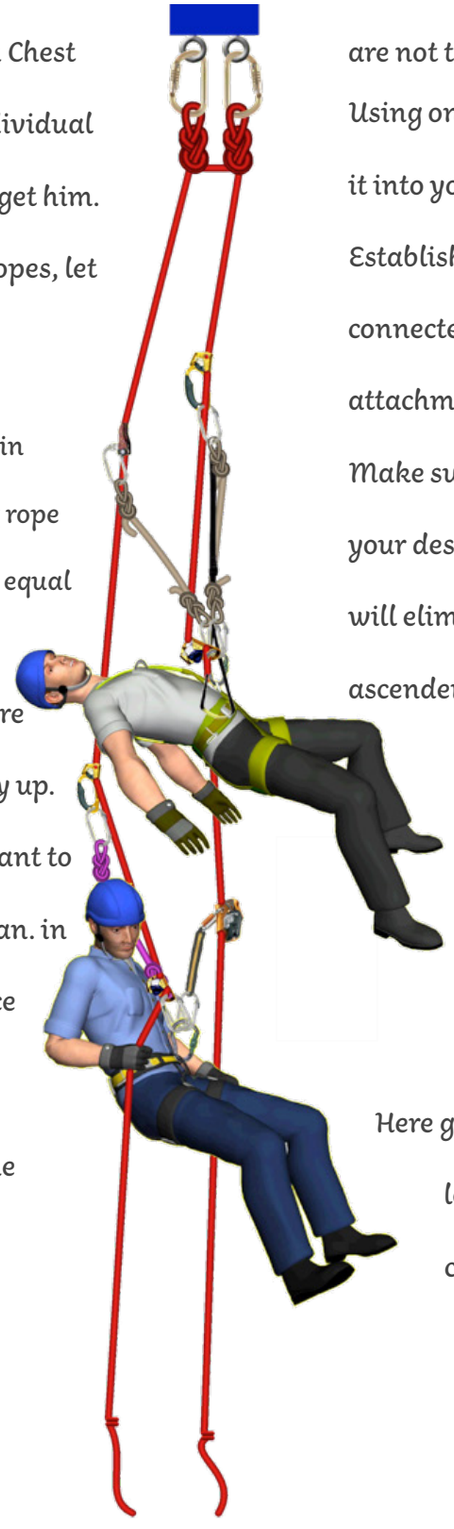
Pick Off - Casualty Ascending

Similar to the method used for a pick-off from the descender, pick-off from Chest Ascender will start by you assessing the condition of the incapacitated individual and ensuring that you will not be placing yourself in harm's way if you go get him. That's pretty important as it's hard enough to rescue one person from the ropes, let alone two or more.

To begin, the actual rescue (assuming that you've ensured that you will be safe in doing the rescue), connect your back-up safety device to the casualty's working rope and your ascenders to the casualty's safety rope. This will allow approximately equal loading of both ropes during the ascent.

Ascend to the casualty until you are level with the casualty. Place your spare back-up safety device above the casualty's ascender and push it all the way up. The casualty's hand ascender may be in the way, push it up, too. It's important to keep your back-up device lower than the casualty's hand ascender if you can. in all cases remember that a back-up device should have about 4" of clearance above a toothed device. Remove your first backup device from the rope.

Ascend past the casualty and change over into your descender. Lock off the descender. Make sure the ropes



are not trapped between you and the casualty, move them to one side..

Using one of the casualty's cow's tails, establish a link between the two of you by clipping it into your ventral attachment. This establishes one point of contact to the casualty.

Establish the second point of contact by connecting a short link or two carabiners connected to each other from your descender carabiner to the casualty's sternal attachment point.

Make sure you're high enough above the casualty that the short link points down from your descender carabiner to the sternal attachment on the casualty's harness. Doing this will eliminate the need to lower the casualty so much once you take off the chest ascender.

Now it get's fun. The challenge here is that you have to lift the casualty enough to create slack above his chest ascender so that you can remove it.

There are many ways to accomplish this maneuver. We'll just talk about one here, the way we teach this.

Here goes, you'll need a few extra bits of gear; one rope grab, one small pulley and a length of accessory cord, at least 6' long. It might be handy to have a few spare carabiners too.

You are positioned above the casualty with two points of contact to the casualty. You're also in the descender to make it easy to descend later. Confirm all of this and then remove the casualty's hand ascender and safety back-up. Next, attach a rope grab to the casualty's working line, above his chest ascender. Add the pulley to the rope grab with a carabiner, making sure the pulley is closed, if it has moving side plates, and the carabiner is locked. Using the end of the accessory cord, attach it to the carabiner holding the pulley with an appropriate knot. Run the cord through the casualty's central attachment, through the pulley and repeat twice so that you've created a 4:1 mechanical advantage system.

Pull down on the cord to remove all of the slack and stretch between the casualty's harness and the pulley/rope grab combination. When all of the slack is taken out of the system, and only then, put the hand ascender on to the cord, a bit lower than the pulley. Put your foot into a shortened foot-loop.

Check the position of your back-up device and ensure that it is as high as practicable, but not tight.

Reach down and grab the casualty's harness, preferably at the waist/ventral attachment. While pulling up with your arms, push down with your foot. This action will lift the casualty enough to remove the chest ascender from the rope.

You need to lower the casualty onto the short link. Squeeze the cord at the bottom of the pulley to apply friction to your mechanical system. Take the hand ascender off of the cord while this friction is held tightly. Safely lower the casualty until the system is slack. Take your lifting system off of the line and off of the casualty.

Place an extra carabiner on the carabiner connecting your descender to your harness, on the side of the descender facing you. Feed the slack end of the rope coming from the descender through this carabiner. This will act as extra friction while descending allowing you to control the descent.

It's worthwhile to note here that you must maintain a small fall factor by keeping your back-up device as high as practical. Never allow the back-up device to be low enough to create a fall factor greater than .5.

Unlock your descender, pulling the slack end of the working line upwards so the carabiner you installed applies friction. Descend until you are near the ground maintaining that the casualty does not fully lie on the ground. It is imperative that all of the weight is not taken off of the casualty all at once as, if the casualty has been suspended without movement for a prolonged period of time, the sudden release of pressure could send too much blood surging through the arteries and cause additional injury. Slowly lower the casualty to the ground all the while moving the legs to restore circulation.

SPRAT Level 1 Rescue Scenario

SPRAT requires Level 1 candidates to perform a "rescue scenario". Essentially it asks the candidate to perform the following:

The casualty will be on a rope system, in his ascending equipment. The rescuer will approach the casualty from an adjacent set of ropes. Once in position, the rescuer will install the casualty's descender directly under the casualty's chest ascender on the working line.

The rescuer will then perform a pick off from ascent and lower the casualty onto the casualty's descender and then perform a rescue from descent.

Not much to it, really, just combining two rescues into one. We've broken the rescues into two separate exercises so that you'll understand how each is performed and because IRATA requires them to be separate.

Anchorage Systems

There are a few anchor systems you'll need to learn as a Level 1 Candidate:

Simple Structural Anchorage

In it's simplest form, an anchorage is made up of two anchors connected around an anchorage, each with a rope attached to the connecting carabiner. You can take advantage of load sharing capabilities by passing the bights of each knot through each connector so the anchors remain separate but the bight of the rope you're hanging on will load both of the connectors. Make sure you tie end knots before you get on the ropes.

Load Sharing Anchorage

You'll have to demonstrate installing load sharing anchors with two anchorages less than 1 meter apart. We often install load sharing or "Y-Hang" systems when we want to avoid hazards such as edges or hot surfaces. In a "Y"-Hang, a Figure "9" is tied near the end of a rope. There should be enough of a tail on the Figure "9" so that the end falls at least 3' below the "y". Make sure you tie an end knot on the tail. Tie an Alpine Butterfly on the other rope coming from the Figure "9". The Alpine Butterfly is tied with enough slack in the loop to allow load sharing between two offset anchor slings. Don't make the included angle go over 120°; however, 90° would be better.

Alternately, a Double Figure "8" on the Bight (Rabbit) Knot can be tied with long enough ears to keep the angle loading down to a minimum.

There are a few considerations you need to consider when you're installing this type of system:

Consequences of failure - Where will the load end up in the event one side of the system fails? How do we minimize the results?

Anchorage location - Are the anchorages suitable for use? Do we need to protect any of the system components while in use?

Included angle loading - we have to consider the included angle and how the resultant forces affect the system components.

Connector loading - Are the anchors and connectors being loaded properly?

Edge Protection - Will our system require any additional edge protection? This can be a problem if we are using the Y-hang to avoid an edge and we have a system failure which results in ropes coming into contact with the edge we are trying to protect.

Rigging and Operating a Hauling System

You'll be asked to put together a simple hauling and lowering system and operate it. This is pretty basic stuff, but it can get a little confusing to begin with. Once you understand how to put these systems together, you'll find you use them all the time. In fact, this system is the basis for our other rigging systems.

A lift / lower system really has three components: a lowering system, a back-up system and a lifting system.

Begin with selecting overhead anchorages above the load. You'll install pulleys here so that you can lift the load straight up. You want the anchorages to be a near to overhead as possible to limit the potential swing or movement of the load once you begin lifting. This is a good time to determine where you'll operate the equipment, too. Select an area away from the load where you can keep an eye on the load, the load path, and the entire rope system.

Install pulleys at the designated anchorages and run ropes through these pulleys. One set of ends will be attached to the load, so make sure they are aligned properly coming out of the pulleys and are long enough to reach the load. Tie a slip-knot on both sides of the pulleys so they won't go whizzing past you and fall to the ground.

The ropes from the other side of the pulleys will be taken over to the controls and loaded into the descender and a safety back-up. If the load is heavy, use two descenders. Also, make sure the ropes are not crossed or tangled along the path from your controls to the pulleys.

If you've done this correctly, you've installed the first two components of your hauling system. the descender is used to lower the load, the back-up serves as, well, a back-up. You operate this equipment just like the equipment on your harness. Now you need to install a simple lifting system.

To install a mechanical advantage system, you'll need a rope grab such as a Croll, Basic, Shunt, Duck or other device that will simply grab a rope without slipping under lower static forces. I prefer a Shunt. Attach the shunt on the line leading from the descender to the pulleys. This is the line connected to your load. Once installed, grab the rope on the other side of the descender (the standing line) and place it through a personal pulley. Attach this pulley on the rope grab installed earlier. Presto-chango! you've installed your first mechanical advantage system.

I might explain a bit about how mechanical advantage works. MA is created when a moving pulley is attached to a system supporting a load. When used, the pulley reduces the effort needed to lift the load. This is achieved through the efficient use of amplification of the input forces by introducing movement. So, you're hand pulling on a rope is the input force and the pulley that moves amplifies the force resulting in increased output force, which moves the load. Ah, the mysteries of physics! I bet you weren't expecting a science lesson.

The amount of mechanical advantage we've created is expressed as a ratio, 3:1, meaning every unit of force you place on the pull rope will result in three times the force placed on the load. We usually just divide the total weight of the load by the MA ratio to determine the amount of effort needed to lift the load. So, if the load weighs 100 kg, then we need 33.33 kg of effort on the pull rope to move the load up.

Ok, our system is installed and ready to lift. Make sure your load and load path are clear and you can monitor them throughout the operation. Unlock the descender and begin to lift the load slowly by pulling on the standing end of the rope. Pulling slowly ensures the load does not experience an unexpected or uncontrolled shift. The load will want to stabilize, or come to rest, below the overhead anchors. Once stabilized, continue to pull the rope to lift the load. You'll notice the rope grab and pulley are getting closer to the descender with every pull. You'll have to lock the descender and reposition the rope grab further up the rope every once in a while.

Now it's time to lower the load. Remove your lifting system. We believe it is always necessary to add additional friction to the standing end of the rope before you lower with a descender. Doing this will allow better control over the load. Reposition the backup device so that there is a bit of slack in the lanyard. Remember, you cannot have a fall factor greater than 1. Using additional friction, operate the descender to lower the load. Stop operating the descender and give yourself more slack on the

back-up as needed. Repeat as necessary until the load is safely on the ground.

It's most important in lifting and lowering operations that you watch the load and the load path, not the equipment that you're using. As the back-up line begins to become slacked, mind the back-up device as appropriate.